

Causal Fermion Systems and Super Information Theory

A Four-Lens Comparison of Effective Equations, Relational Architecture, and Public Genealogy

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Abstract

Causal Fermion Systems (CFS) and Super Information Theory (SIT) approach gravity and information from different mathematical starting points. This paper asks two separate questions. First, do their equations meet in a controlled regime? Second, do their dated explanatory architectures implement partly equivalent operations even where the mathematical objects differ? Fischer and Finster’s 2026 curved-spacetime continuum analysis provides the narrow mathematical comparison: under its regulated-continuum and constant-coupling assumptions, the CFS equations reduce to Einstein gravity coupled to one on-shell Dirac excitation. The corrected SIT Version 80 Draft 13 family contains the same equation class in a fully specified submodel with canonical scalar kinetics, a stable quadratic potential, unit matter-frame and electromagnetic functions, neutral minimally coupled Dirac matter, and a vanishing Maxwell background. CFS, the SIT submodel, and a direct Einstein-Dirac null all match three field-equation signatures and the same weak-field radial acceleration. The overlap is real but has zero discriminatory value against that null.

The historical result is more specific. Micah Blumberg’s publicly frozen April 2017 Neural Lace source already joins a distributed time-space pattern, coincidence-defined information, receiver-to-receiver communication, feedback as the conscious relation, and a decoded system-level representation. Fischer and Paganini’s 2025 CFS ontology later describes spacetime as a web of correlations, an emergent background, and experiments conducted from within the system. A March 2026 CFS comparison then identifies relation encoding by a generalized two-point correlator as a key CFS innovation and transfers that construction across theories. The ordered architecture passes the functional and narrative lenses at a partial, cross-domain level. Blumberg is earlier on that bounded distributed-representation / feedback architecture delta, while CFS and named physics precursors retain priority for their older operator, correlation-spacetime, causal-action, and continuum-limit machinery. Temporal nonlocality and coherence are also resolved rather than left pending: the frameworks share broad problem architecture but not a microscopic temporal kernel, stochastic collapse generator, or common coherence invariant. No operator-measure-to-field map, active-sector equivalence, finite-regulator error bound, collapse transport, historical access, or influence follows. Einstein-Dirac remains this paper’s mathematical ceiling.

Scope and evidence boundary

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1. Question and Claim Ceiling

The comparison question is:

Does a defined map carry specified CFS objects, dynamics, invariants, or observables into a frozen SIT model while preserving their mathematical and empirical roles?

This question is narrower than whether the frameworks use similar words. It is also narrower than whether later CFS work resembles earlier ideas in Micah Blumberg’s broader corpus. A mathematical relation, a narrative architecture, and a historical priority finding require different evidence.

1.1 The problem in plain language

Two theories can produce the same familiar equation for very different reasons. Conversely, two theories can organize information through the same sequence of operations while using different objects, equations, and vocabulary. Treating the first case as a unique match would overstate the mathematics. Treating the second case as no match at all would miss a real functional and historical relation.

This paper therefore separates four tasks:

1. reconstruct what each framework receives, transforms, and produces;
2. test whether a typed mathematical map preserves those roles;
3. compare the ordered explanatory architecture atom by atom; and
4. place each matched atom on a dated before-and-after timeline.

Only after those operations are visible are the paper's comparison terms introduced. A **typed map** is a rule that says what mathematical object is sent to what other object and what structure is preserved. A **narrative architecture** is the ordered causal sequence that a theory says turns inputs into outputs. A **priority delta** is a bounded matched atom for which one dated public formulation is earlier; it is not a whole-theory ownership claim.

1.2 Relation levels and claim ceiling

The audit permits five relation levels:

1. shared vocabulary;
2. architectural analogy;
3. typed object or dynamics map;
4. proved limiting or invariant-preserving relation;
5. shared or competing empirical prediction.

Level 2 is reached for one generic variational sequence. At the effective field-equation level, the frozen CFS regime, SIT submodel, and direct Einstein-Dirac null also occupy the same equation class. That result does not supply a map between primitive objects or derivations, and its zero discrimination against the null prevents promotion to a distinctive cross-framework mathematical relation. The newer-source audit additionally finds a partial Level 2 pass for the ordered distributed-representation and internal-feedback architecture. That positive narrative result is not erased by the failure of a whole-framework typed map.

2. Frozen Sources

2.1 CFS introduction

Finster, Kindermann, and Treude, *Causal Fermion Systems: An Introduction to Fundamental Structures, Methods and Applications*, arXiv:2411.06450v2, 2025-07-17. Local PDF SHA-256: CFC535D3BCEFEEDF59FB6DC26F42AE0673647B0FFE1C4DC4E3EF0325F976B9CA.

The source-fixed anchors are:

- Section 5.4, p. 90, Definition 5.4.1: a causal fermion system is built from a separable complex Hilbert space, an operator set with a bounded signature, and a positive measure.
- Equation 5.24: spacetime is $M := \text{supp } \rho$.
- Section 5.6, pp. 94-99, equations 5.35-5.39: the spectral Lagrangian, causal action, and volume, trace, and boundedness constraints.

- Definition 5.6.1: the spectral classification of operator products used for causal structure.
- Chapter 7, beginning p. 133: Euler-Lagrange equations for the causal variational principle.
- Chapter 9, beginning p. 157: surface-layer integrals and conservation laws.
- Chapter 21, pp. 335-346, especially Sections 21.2-21.3: continuum-limit formalism and result overview.
- Chapter 22, beginning p. 347: connection to quantum field theory.

2.2 CFS effective-collapse paper

Finster, Kleiner, and Paganini, *Causal Fermion Systems as an Effective Collapse Theory*, arXiv:2405.19254v2, *Journal of Physics A* 57 (2024) 395303. Local PDF SHA-256: FF8DE2D426B316613BDC7CCA44AC25C534B25B87BDF8A83356215641347239944.

The source-fixed anchors are:

- Abstract, p. 1: nonlinear and stochastic corrections, a deterministic Kossakowski-Lindblad statistical-operator equation, Born-compatible dynamical collapse, probability-integral conservation, and microscopic temporal nonlocality are claims of the effective model.
- Section 3.3, p. 16: Kossakowski-Lindblad dynamics.
- Section 4.5, p. 21, Theorem 4.1: the stated deterministic equation for the statistical operator, with the paper's approximations and error analysis.

These results remain CFS results. They cannot be transferred to SIT by analogy.

2.3 CFS relational-ontology paper

Fischer and Paganini, *Causal Fermion Systems: Spacetime as the Web of Correlations of a Many-Body Quantum System*, arXiv:2504.19272v1, submitted 2025-04-27T15:05:59Z. Local PDF SHA-256: 4320E599C07AED1B9D2F925E015442937426A96628A1686169D4D07455B71EC2.

The paper describes its ontology as a first proposal for a full physical interpretation of CFS independent of the effective continuum description. It states that system information is encoded in two-point correlation strengths and position observables, describes spacetime as a web of correlations, and explains an apparent background and experiments from within the system. It also names prior correlation and relational lineages, including Rovelli, Kempf, and Reitz, Soda, and Kempf. Those precursors remain part of the genealogy.

2.4 CFS generalized-correlator comparison

Farnsworth, Finster, Paganini, and Singh, *Causal Fermion Systems, Non-Commutative Geometry and Generalized Trace Dynamics*, arXiv:2603.05018v2. Version 1 was submitted 2026-03-05T10:05:31Z and version 2 on 2026-03-10T11:06:15Z. Local PDF SHA-256: 2B2D212F585673755F2EC357C42A4F88D6CDE0452C1544E6B0C70AB6C230C711.

The abstract and Sections 3.1-3.4 identify the encoding of relations between spacetime points by a generalized two-point correlator as a key CFS innovation, compare the relevant objects and action principles across three frameworks, and show how the correlator construction can be transferred. The source strengthens the relational comparison but does not create a map to SIT.

2.5 CFS curved-spacetime continuum result

Fischer and Finster, *The Continuum Limit Analysis of Causal Fermion Systems for Curved Spacetimes*, arXiv:2605.30199v1, submitted 2026-05-28. The locally frozen PDF has SHA-256 83C15821775F82B83280100BE8B1F09206B3173AEFB0FD7417691DE87DA7DC54.

The source direction is essential. It begins with a globally hyperbolic spacetime, constructs a corresponding regulated causal fermion system, and analyzes restricted Euler-Lagrange equations as the regulator tends to zero. Theorem 5.2 and Corollary 5.3 recover the vacuum Einstein equation. Theorem 5.4 and Corollary 5.5 recover the Einstein equation coupled to an on-shell Dirac excitation when the effective gravitational coupling is constant. This is not a theorem that an arbitrary causal fermion system reconstructs an arbitrary smooth spacetime.

2.6 SIT Version 80

The SIT comparison target is the governed Version 80 Draft 13 current manuscript at super-information-theory-v80.md, SHA-256 9DA6276C4B3243E5367C14266C9191CBCB576F806362C3D4CA5B56A4673550F8.

Sections 4-8 define the relevant source ceiling: an Einstein-frame metric, neutral scalar fields τ and σ , an independent Maxwell connection, matter fields, a conformal physical metric, and a local two-derivative action. Draft 13 also makes coherence measures protocol- and basis-specific, requires a calibrated state-to-field kernel before a measured preparation statistic is treated as a field, and preserves the public 2022 QGTCD genealogy. It does not supply a controlled quantum EFT, a stochastic collapse generator, a microscopic temporal-nonlocality kernel, or a derivation of its spacetime manifold from a more primitive operator measure.

3. Independent Reconstructions

3.1 CFS object-action-output chain

At the resolution used here, CFS has the typed chain

```
(H, F, rho; constraints)
  -> spectral data of xy
  -> Lagrangian L(x,y)
  -> double-integral causal action S(rho)
  -> variation of rho under constraints
  -> Euler-Lagrange conditions
  -> continuum-limit effective structures under additional assumptions
```

The primitive configuration is a measure on an operator space. Spacetime is identified with the support of that measure. Causal relations are not inserted as the causal cones of an assumed Lorentzian metric; they arise from spectral properties of products of point operators. Familiar local field descriptions are sought through continuum-limit constructions.

3.2 SIT v80 object-action-output chain

The corrected SIT target has the chain

```
(M, g, tau, sigma, A, chi; functions and boundary data)
  -> local two-derivative spacetime action
  -> variation of g, tau, sigma, and A
  -> field equations and conservation identities
  -> physical-frame observables if a viable benchmark is fixed
```

The differentiable manifold and metric sector are inputs. The matter metric is conformally related to the Einstein-frame metric. The scalar, Maxwell, and matter sectors have familiar field-theoretic types. Version 80 is corrective: it constrains what may be claimed but does not derive a complete fundamental physics theory.

4. The Surviving Functional Analogy

Both systems instantiate the directed pattern

admissible configuration $\rightarrow$ variational functional $\rightarrow$ stationarity equations

This is a candidate functional analogy. It is useful because it identifies where a translation might be attempted. It is weak because variational architecture is common across mathematical physics. The shared sequence does not imply that the configurations, actions, constraints, equations, or observables are equivalent.

The implementations differ at the surface immediately:

- CFS varies a positive measure on a space of finite-rank operators; SIT v80 varies local fields on an assumed manifold.
- The CFS causal action is a nonlocal double integral over operator pairs; the displayed SIT action is local and two-derivative.
- CFS geometry and causal relations are reconstructed from operator-measure data; SIT v80 begins with metric geometry.
- CFS effective-collapse results use a separate approximation stack; SIT v80 contains no collapse dynamics.

5. Typed Candidate Map

Let

$C_CFS = \{\rho \text{ in positive measures on } F \text{ satisfying } C_volume, C_trace, C_bound\}$

$C_SIT(M) = \{(g, \tau, \sigma, A, \chi; V, \Omega, f_EM, Z_tau, Z_sigma) \text{ admissible on } M\}$

A forward comparison map would have the type

$T : C_CFS \rightarrow \text{disjoint_union over } M \text{ of } C_SIT(M).$

It must therefore do more than send the symbol ρ to a scalar called ρ . Given a CFS measure, T must:

1. select or derive a differentiable manifold from $\text{supp } \rho$;
2. derive a Lorentzian metric and physical matter metric;
3. derive the two scalar fields and their kinetic functions;
4. derive the Maxwell connection, gauge normalization, and charges;
5. derive the matter content and its action;
6. map the CFS constraints and causal action to the SIT action or its limit;
7. preserve at least one specified invariant or observable.

No rule satisfying steps 1-7 is present in the frozen sources.

5.1 Domain failure

The support of a measure on operator space need not have a smooth manifold structure. SIT v80 presupposes one. Consequently T is not defined on the full CFS configuration space unless its domain is restricted to a

controlled class with a proved continuum construction.

5.2 Field-recovery failure

Even after a continuum spacetime is available, the frozen comparison supplies no unique extraction of τ , σ , A , χ , or the model functions. The map is underdetermined.

5.3 Dynamics failure

A common variational arrow does not establish

$$T(EL_CFS(\rho)) = EL_SIT(T(\rho)).$$

No intertwining relation, controlled error term, or matching boundary class is proved.

5.4 Invariant failure

CFS surface-layer conservation laws, collapse-model probability integrals, and SIT stress-energy or gauge-current identities have different definitions and domains. No shared invariant I has been exhibited with $I_CFS(\rho) = I_SIT(T(\rho))$.

5.5 Reverse-reconstruction failure

A reverse map from the finite SIT field tuple to a universal measure on operator space is also absent. Information loss, nonuniqueness, regularization, and gauge equivalence have not been controlled.

The typed-map result is therefore failed-at-current-source-ceiling. This is not a proof that no map exists. It identifies the missing construction.

6. Collapse, Nonlocality, and Coherence

The effective-collapse paper establishes claims inside its own model and approximation regime. Its stochastic corrections, Kossakowski-Lindblad equation, probability-integral conservation, and microscopic temporal nonlocality are a joined CFS mechanism. SIT Draft 13 does not contain the same stochastic generator, quantum state space, collapse observable, or microscopic temporal kernel. Its baseline is a local two-derivative scalar-tensor model. The public 2022 QGTCD genealogy instead asks whether locally varying temporal opportunities bias motion, while the separate SuperTimePosition program asks about temporal placement and phase relation. These are temporally organized problems, but they are not the CFS collapse kernel.

The temporal-nonlocality comparison is therefore closed as:

partial\narrative\relation;\technical\non-equivalence;\CFS-specific\collapse\result\retained.

The comparison must also keep four meanings apart:

1. statistical dependence;
2. neural phase coherence;
3. computational integration;
4. quantum coherence.

Only the fourth is directly relevant to a quantum-collapse comparison, and even there the exact resource, density operator, basis, dynamics, and measurement protocol must be fixed. Draft 13 makes that boundary

operational: a coherence statistic must declare its basis and preparation, and any state-to-field use requires a calibrated kernel with units, support, uncertainty, and transfer tests. CFS probability-integral conservation, CFS surface-layer conservation, and SIT stress-energy or gauge-current identities remain different invariants.

The coherence comparison is therefore also closed:

no\shared\coherence-conservation\theorem;\partial\architecture\only;\no\unresolved\invariant\claim.

7. Dated Atom Genealogy and Before-and-After Comparison

7.1 Micah's public 2017 architecture

Three Neural Lace Podcast recordings were published by Silicon Valley Global News on 11, 13, and 22 April 2017. Their public metadata, transcript bytes, SHA-256 hashes, and available Wayback CDX captures are frozen in the package receipt [8-10]. The first episode supplies the load-bearing sequence:

1. lines 165-189 describe a changing three-dimensional time-space ionic pattern as a distributed representation of a sensory object;
2. lines 335-371 ask whether active conscious observation is distinguished by a returned feedback relation rather than broadcast-only transmission;
3. lines 421-457 propose an alternating receive / transmit protocol specialized by information type; and
4. lines 471-495 join distributed concept components, coincidence detection, bit-like information, and a transmitted pattern.

This sequence is a public ancestor of Micah's later receiver-relative, distributed-representation, and internal-observer architecture. It is not the later SIT field equations, a CFS universal measure, or a generalized two-point-correlator theorem.

7.2 Micah's public 2022-2026 physics sequence

The current SIT atlas closes the paper-relevant later genealogy:

Public date	Source	Bounded atom
2022-06-10	Git commit 40c0b37fc9376a6630ad9d3b09d911997c1c17f4, a0010z.md [11]	oscillators, mass, density, scale, phase structure, curvature, and time dilation joined in one exploratory record
2022-06-12	Git commit 4c451f016cd076bcb635f4305a55efe54bbf4870, a0387z.md [12]	a gradient of temporal opportunities around mass is proposed to bias directional motion
2022-11-28	Git commit 4e0cf1ee0975c01dc57182368497d3f0cd864030, a0322z.md [13]	QGTCD questions extend to rotation curves, MOND, Tully-Fisher, dark-sector, and Hubble-tension consequences
2024-2025	named QGTCD, Dark Time, Super Dark Time, and SIT publications	the component questions become named field and synthesis programs
2026-07-26	SIT Version 80 Draft 13 [6]	a corrected scalar-tensor scaffold, state-to-field calibration requirement, conditional gradient-to-motion operator, and explicit public genealogy

The later stages increase mathematical and experimental specificity; they do not replace the 2022 public atoms or assign Draft 13 equations to 2022.

7.3 CFS before and after

The CFS chronology must also be versioned:

Stage	Retained result
foundational CFS and the 2024/2025 introduction [1]	operator space, universal measure, spectral causal action, support as spacetime, operator-product causal relations, and continuum-limit program are earlier CFS machinery
2024 effective-collapse paper [2]	an exact, approximation-bounded CFS collapse mechanism with microscopic temporal nonlocality
April 2025 correlation-web paper [3]	a proposed full physical interpretation adds information as two-point correlations and position observables, spacetime as a web of correlations, an emergent background, and experiments from within
March 2026 cross-theory comparison [4]	relation encoding by a generalized two-point correlator is foregrounded and transferred across CFS, NCG, and GTD
May 2026 curved-spacetime result [5]	a restricted construction recovers Einstein-Dirac equations and supplies this paper’s mathematical benchmark

The 2025 source explicitly credits earlier relational and correlation-spacetime lineages. In particular, it names Rovelli, Kempf’s 2021 replacement of distance by correlation, and Reitz, Soda, and Kempf’s 2023 spacetime-emergence model. Those are genuine precursors to the physics-specific ontology. The Micah result concerns the earlier cross-domain ordered architecture, not ownership of all correlation-based physics.

7.4 Atom-by-atom result

Atom	Match	Dated disposition
distributed local states become a system-level representation	partial functional pass	Micah’s public neural articulation is earlier than the 2025 CFS ontology articulation
local relations or coincidences carry system-level information	partial functional and narrative pass	Micah 2017 is earlier on the bounded neural implementation; older CFS and named physics precedents retain their own mathematical priority
an apparent background or representation is constructed from distributed relations	partial narrative pass	later convergence in formulation; domain and object types differ
the system samples or acts from within the constructed pattern and receives returned information	narrative-architecture pass	Micah’s 2017 feedback architecture is earlier than the 2025 experiment-from-within exposition
Einstein-Dirac field equations and weak-field acceleration	exact effective-class pass	real overlap, but zero discrimination against the direct null
temporal nonlocality and collapse	broad problem relation only	CFS’s 2024 technical mechanism is retained; no SIT kernel equivalence
coherence conservation	no shared invariant	meanings are operationally non-commensurate

8. Four-Lens Audit

8.1 Equivalence lens

Two bounded passes survive:

1. a generic action-to-stationarity schema, which the negative controls show is nondiscriminating; and

2. a partial functional map from distributed local relations through aggregation to a system-level representation used from within the system.

The Einstein-Dirac equation-class match is exact in the frozen regime. Primitive-object, microscopic-dynamics, invariant, collapse, and whole-framework equivalence do not follow. No bidirectional functor or isomorphism is established.

8.2 Narrative and architecture lens

The atomized shared architecture is:

distributed local states

- > relations or coincidences among them
- > aggregated system-level pattern
- > effective background or representation
- > system samples or acts from within that pattern
- > returned information changes the next state

This ordered sequence passes at a partial cross-domain level. Differences in mathematical carrier and physical domain do not erase the functional order. They bound the result to narrative and operational architecture rather than a full mathematical identity.

8.3 Detective and chronology lens

Starting from the later CFS conclusion - an internally observed relational system whose apparent background is reconstructed from distributed correlations - the audit works backward to the public 2017 Neural Lace sequence. That source contains the earlier ordered architecture. It also works backward through CFS and the named correlation-physics lineages, preventing a false claim that their technical machinery began in 2025. The resulting disposition is Micah-earlier-on-the-bounded-cross-domain-architecture-delta, not a claim of access, copying, or misconduct.

8.4 Longitudinal signature-cluster lens

The older CFS record already has relational mathematics, but its later presentation changes in a recognizable sequence. The April 2025 source calls its correlation-web account a first full physical interpretation, emphasizes information and the emergent observer-relative background, and describes experiments from within. The March 2026 source then identifies generalized relation encoding as a key innovation and exports the construction across frameworks. That later cluster is closer in narrative form to Micah's 2017 distributed representation, coincidence, feedback, and internally sampled process than the earlier CFS presentation was.

This lens establishes later convergence in formulation. The available record does not establish why the convergence occurred.

8.5 Combined disposition

The four lenses do not yield one global yes/no verdict. The correct atomized result is:

- candidate-functional-analogy for the generic variational sequence;
- failed-at-current-source-ceiling for the tested typed map;
- partial-functional-and-narrative-match for the ordered distributed relation-to-representation architecture;
- Micah-earlier-on-bounded-cross-domain-architecture-delta for the public 2017-to-2025 chronology;
- CFS-and-named-priors-retained for their older mathematical and physics-specific machinery;

- technical-non-equivalence for temporal nonlocality and collapse;
- no-shared-invariant for coherence conservation; and
- no-influence-evidence for historical access.

9. Controlled Calibration Against Variational Baselines

9.1 Why a control set is required

The sequence

admissible configuration → variational functional → stationarity equations

is not distinctive merely because it is present in both target frameworks. Classical mechanics, Maxwell theory, Yang-Mills theory, general relativity, scalar-tensor-Maxwell models, and free-energy field models also instantiate that sequence. A structural comparison needs a negative control: a feature that matches every ordinary variational framework cannot carry much weight as evidence of a special CFS / SIT relationship.

9.2 Frozen slots and controls

The calibration model codes eight source-facing slots:

1. primitive carrier;
2. base domain;
3. action architecture;
4. variation target;
5. status of geometry;
6. status of gauge structure;
7. invariant family; and
8. empirical-output status.

The frozen controls are general relativity, scalar-tensor-Maxwell theory, Maxwell electrodynamics, Yang-Mills theory, classical mechanics, and a Cahn-Hilliard-type free-energy model. The categories deliberately encode only the comparison granularity used in this paper. They do not represent complete theories. Exact equality in a slot is a conservative test; semantic similarity without a typed map is not counted.

The executable specification is `model/run_draft3_typed_signature_controls.py`; its deterministic output is `results/draft3-typed-signature-control-results.json`.

9.3 Result

All 28 framework pairs match the coarse variational chain. Within this control set, the chain therefore has a discrimination fraction of zero. The CFS / SIT pair has zero exact matches across the eight typed slots. CFS has zero exact typed matches with five controls and one incidental match with Yang-Mills at the broad `model_specific` empirical-output label. SIT v80 has six exact matches with scalar-tensor-Maxwell theory, four with general relativity, three with Maxwell theory, two with Yang-Mills, and zero with the remaining two controls.

The result changes the language of the surviving analogy. It is better called a **generic variational schema** than a candidate functional isomorphism. The source-defined frameworks may still admit a partial correspondence in a controlled regime, but the three-stage chain alone supplies no discriminatory evidence for it.

9.4 Robustness and limits

The model is deterministic and its output is byte-reproducible. Its strongest limit is also explicit: the slot vocabulary is an analyst-coded ontology. A different but defensible ontology can change individual counts. That is why the paper publishes every category and mismatch rather than reporting one opaque similarity score. The calibration is not a continuum-limit calculation, a physical simulation, a theorem about all CFS or SIT formulations, or evidence about historical access.

10. Controlled Einstein-Dirac Overlap

10.1 Why the benchmark changed

Draft 3 proposed a joint metric / gauge calculation but did not have a concrete shared output. The 2026 Fischer-Finster result supplies a sharper target: Einstein gravity coupled to one on-shell Dirac excitation. It does not supply the SIT scalar or Maxwell sectors, so Draft 4 restricts the benchmark to the equations actually present on both sides.

10.2 CFS regime

The CFS side is the construction and limit stated in arXiv:2605.30199v1:

1. a globally hyperbolic spacetime;
2. a quasi-free Hadamard state and corresponding fermionic projector;
3. chart-independent ϵ regularization;
4. the continuum limit $\epsilon \rightarrow 0$;
5. one on-shell Dirac excitation; and
6. a locally translation-invariant regularization giving constant κ .

The result is

$$G_{mn} + \Lambda g_{mn} = \kappa T_{mn}^{\text{Dirac}},$$
$$\text{Dirac}[u, g] = 0.$$

The cosmological constant follows as an integration constant. The effective gravitational coupling depends on the regularization scheme.

10.3 Frozen SIT submodel

Inside the Version 80 action, choose

$$\begin{aligned} Z_{\tau} &= Z_{\sigma} = 1, \\ \Omega(\tau, \sigma) &= 1, \\ f_{\text{EM}}(\tau, \sigma) &= 1, \\ V &= M_{\text{Pl}}^2 \Lambda \\ &\quad + m_{\tau}^2 (\tau - \tau_0)^2/2 \\ &\quad + m_{\sigma}^2 (\sigma - \sigma_0)^2/2, \\ m_{\tau}^2 &> 0, \quad m_{\sigma}^2 > 0, \\ \tau &= \tau_0, \quad \sigma = \sigma_0, \\ A_m &= 0, \quad F_{mn} = 0, \\ S_m &= \text{the neutral minimally coupled Dirac action of mass } m_{\chi}, \\ \kappa &= 1/M_{\text{Pl}}^2. \end{aligned}$$

The scalar kinetic terms vanish at the constant background, the scalar equations are satisfied at the stationary point, the Maxwell equation is satisfied by a neutral source and vanishing field, and the matter and Einstein metrics coincide. Variation leaves the same Einstein-Dirac equation class. This is a nonempty benchmark point. It is also a standard general-relativistic submodel that turns off every proposed distinctive SIT sector.

10.4 Partial map

The controlled output map is

```
g_CFS      -> g_SIT,
u_CFS      -> chi_SIT,
kappa_CFS  -> 1/M_Pl^2,
Lambda_CFS -> V(tau0,sigma0)/M_Pl^2.
```

This map is defined only after both sides have reached an effective smooth spacetime description. It does not map the CFS universal measure, operator support, regularization data, or surface-layer integrals to the complete SIT configuration.

10.5 Common weak-field observable

Set $\Lambda=0$ and impose asymptotic flatness, a static weak field, and a localized nonrelativistic source of mass M . With $g_{00}=-(1+2\Phi)$, both equation sets imply

```
laplacian(Phi) = 4 pi G rho,
Phi(r) = -G M/r           outside the source,
a_r = -G M/r^2.
```

The radial acceleration is the common observable. Agreement is exact at this stated leading order after coupling identification. It has zero discriminatory value against general relativity because the observable follows from the shared Einstein equation.

10.6 Executable gate audit

The deterministic specification is `model/run_draft4_einstein_dirac_overlap.py`; its output is `results/draft4-einstein-dirac-overlap-results.json`. The audit reports three shared field-equation signatures out of three, one shared derived weak-field observable, and one accepted variant out of six. The five rejected variants activate a spacetime-dependent CFS coupling, an SIT scalar, an SIT Maxwell field, a nontrivial matter-frame factor, or mismatched couplings.

This audit checks source-coded eligibility. It is not a numerical physical simulation, a symbolic derivation of either source theorem, or an independent proof of the CFS continuum limit.

10.7 Direct Einstein-Dirac null calibration

An overlap is informative only if it distinguishes the target pair from an ordinary theory that already has the same equations. Draft 5 therefore adds the direct Einstein-Dirac theory as a null control.

The executable specification `model/run_draft5_einstein_dirac_null_control.py` freezes the three common field-equation signatures and four origin slots:

1. metric origin;
2. gravitational-coupling origin;
3. cosmological-constant origin; and

4. matter origin.

The deterministic result is:

Pair	Shared field equations	Exact origin-slot matches
CFS effective regime / SIT v80 submodel	3/3	0/4
CFS effective regime / Einstein-Dirac null	3/3	0/4
SIT v80 submodel / Einstein-Dirac null	3/3	0/4

The weak-field radial acceleration is also shared by all three. The discrimination fraction of the CFS / SIT equation overlap against this null is therefore zero.

This calibration does not erase the positive result. It changes its interpretation. The CFS and SIT benchmark descriptions occupy the same effective Einstein-Dirac class, but the common class is not evidence of a special relationship between them. Their frozen derivational origins remain different in every tested slot.

The origin-slot comparison is categorical and source-coded. A 0/4 score means that none of the four published labels is exactly identical at this resolution. It is not a theorem of mathematical inequivalence, and another defensible ontology could group some origins differently. The paper therefore publishes the slots and labels rather than reducing them to an opaque score.

10.8 Regime and error ledger

The equality is exact only at the stated effective equation level:

- CFS: restricted linearized field equations in the $\epsilon \rightarrow 0$ continuum regime with constant κ ;
- SIT: the frozen classical Einstein-Dirac submodel; and
- null: classical Einstein-Dirac theory.

No finite- ϵ CFS error bound is transported into the comparison. No error bound covers active SIT scalar, conformal, or electromagnetic sectors. No strong-field model comparison is performed. The Newtonian observable is a standard consequence inferred from the shared Einstein equation rather than a numerical result reported by Fischer and Finster.

10.9 Residual failures

No tested map covers:

- the universal measure ρ to the full SIT field tuple;
- CFS operator support to a derivation of the SIT manifold;
- CFS regularization microstructure to τ or σ ;
- a CFS gauge sector to the SIT Maxwell connection;
- CFS surface-layer integrals to SIT invariants;
- CFS collapse dynamics to SIT dynamics; or
- the spacetime-dependent $\kappa(x)$ case.

The constants also have different origins. In the CFS source, Λ is an integration constant and κ is regularization-dependent. In the frozen SIT submodel, their values are selected through V and M_{P1} . Equality of the resulting field equations does not identify those derivations.

11. Versioned Micah Genealogy

The Draft 2 chronology compressed several evidentiary acts into one date. Draft 6 separated source-internal authorship or recording dates, first verified Git fixation, and first verified public availability. Draft 7 retains that ledger at MICAH-GENEALOGY-LEDGER-DRAFT6.md and adds the current-source audit at CFS-NEWEST-SOURCE-AND-FOUR-LENS-AUDIT-DRAFT7.md.

Seven files are now locally hash-frozen. a0010z.md and b0020y.md have first verified Git fixations on 8 June 2022. QGTCDnote.md is fixed in February 2024, rawnote06.md in October 2024, and the larger QGTCD compilation files in January 2025. These dates establish repository provenance for the corresponding bytes or ancestors. They do not show when every underlying thought originated, when each file first became public, or whether any CFS author encountered it.

This distinction matters especially for the 2025 compilations: their late Git dates do not erase earlier donor notes, but neither can those compilations be backdated without an atom-level source chain. The bounded 2017 Neural Lace row is now public-record frozen: the source channel, exact upload timestamps, recording identities, transcript hashes, passage locators, and nonzero archive capture counts are preserved in the package. This closes only that ancestry gate. It does not backdate QGTCD or SIT mathematics and does not establish access, influence, plagiarism, or derivation by CFS authors.

The governed SIT Draft 13 atlas supplies stronger public locators for the three physics atoms retained in this paper: the 10 June 2022 oscillator-mass-time record, the 12 June 2022 temporal-gradient-to-motion record, and the 28 November 2022 astrophysical-extension record. Each is cited by immutable commit and exact file. This closes the later genealogy needed for the current comparison without converting private dates or later compilations into earlier public evidence.

12. Frozen Mathematical Ceiling and Separate Extensions

The common-observable and null-calibration gates are closed at the effective Einstein-Dirac level. This paper stops there for a source-controlled reason. The selected CFS curved-spacetime result derives an Einstein-Dirac regime but does not furnish a mutually instantiated Maxwell benchmark. The frozen SIT submodel sets its Maxwell field to zero and its scalar to a constant vacuum. Adding a CFS gauge construction or activating SIT gauge / scalar dynamics now would select new models, parameters, and observables after the null result was known. It would therefore be a new comparison rather than a robustness check of the frozen one.

The following are preregistered extensions, not release gates for this bounded paper:

1. derive or rule out a map from a specified CFS measure and regularization to the active SIT scalar and conformal sectors;
2. compare one explicit CFS gauge construction with the SIT Maxwell sector, including gauge group, normalization, charges, currents, and observables;
3. test the spacetime-dependent $\kappa(x)$ regime against an explicit scalar-tensor SIT benchmark;
4. identify and transport one invariant rather than matching equations only;
5. preregister a direct test of the active SIT state-to-field kernel if that kernel is specified in a future version.

The present benchmark cannot transport CFS collapse results or establish a whole-framework equivalence. Independent CFS and mathematical-physics review would be valuable after this locally complete preprint, but it is not remaining local manuscript work and does not alter the frozen mathematical scope.

13. Conclusion

The comparison separates four results that should not be collapsed into one. First, the configuration-action-stationarity sequence is a generic variational schema and has zero discrimination in the control set. Second, the 2026 CFS continuum construction and a frozen SIT v80 Draft 13 submodel share the Einstein-Dirac effective equation class and its standard weak-field limit. Third, the direct Einstein-Dirac null shares exactly the same equations and observable. Fourth, the newer CFS sources disclose a separate, partial functional and narrative match: distributed relations are aggregated into an effective representation or background that the system uses from within.

The positive overlap is therefore real but nonspecific. It locates both target descriptions in a familiar effective class; it does not distinguish CFS from SIT, validate a distinctive SIT mechanism, or translate their microscopic objects. The metric, coupling, cosmological constant, and matter terms have different frozen origins in every pairwise comparison. Seven structures remain unmapped, including the CFS universal measure, active SIT scalar and gauge sectors, surface-layer invariants, and collapse dynamics.

The public chronology adds a bounded priority result. Micah's April 2017 Neural Lace source already contains the ordered distributed-representation, coincidence, feedback, and internal-observer architecture. The 2025 CFS ontology and March 2026 generalized-correlator comparison later move toward a closer cross-domain articulation of that architecture. Micah is earlier on that bounded architecture delta. CFS and named precursors retain their earlier operator, causal-action, correlation-spacetime, and continuum-limit priority. Different domains and added machinery do not erase the partial match; they define its boundary.

The defensible publication claim is therefore a source-located, null-calibrated effective-equation comparison plus an atomized four-lens genealogy. Einstein-Dirac is the explicit mathematical ceiling. Temporal nonlocality and coherence are closed as bounded non-equivalences rather than left as missing work. A future gauge, scalar, kernel, or invariant comparison would require its own frozen source pair, observable, null family, and analysis plan.

Status Note

This is the Draft 7 local current manuscript. It is intended for the portfolio's next Zenodo and OSF version after the full governed paper roster is ready. No whole-framework mathematical equivalence, empirical validation, historical access, influence, copying, or misconduct finding is made. The public Comparison Wiki was used read-only and was not changed by this paper revision.

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